# Project 1 Specification: Droning On

*Delivering Solutions Inc. (DS)* is developing a drone-based parcel delivery system with the working title *Droning On*. The system will deliver parcels directly to the home from one of the DS dispatch centres.

Use of drones for delivery is rapidly being pursued by the market. It is important that DS enters the market quickly with a reliable, rapid and safe service which conforms to **Australian Government regulations (https://www.drones.gov.au/)** (link for interest only; no need to read the linked site for this project). To support this, DS has developed a simulation framework to explore the Droning On behaviour.

*Figure 1: One of the DS Drones: DJI FlyCart 30*

**The Current Simulation System**

The system operates with a dispatch centre controlling a fleet of drones delivering parcels to suburban homes. Each drone flies out to the suburb (see Figure 2) from the dispatch centre, entering the suburb at the north end of Out Ave, traversing south to the street to which its parcel is addressed, then east to the street number where it delivers the parcel. It then continues east to Back Ave, and then north before flying back from the suburb to to the dispatch centre. Figure 2 shows the middle of a delivery session with 7 active drones visible including D8 mid-delivery on Fourth Street.

Regulations require that (1) drones only fly above the roads (dark grey in Figure 2) for safety reasons, and that (2) only a single drone at most can be in a rectangular location. However, one drone can enter a location while another exits it. The roads are mapped out in a suburb (represented as a linked structure) to support (1). The dispatch centre manages (2) this by having drones send an access request for a location to which it wants to move. The current system will check if the location is free and, if so, grant access. As such the current system operates on a simple first-come first served basis.

The drones always fly counterclockwise through the suburb to reduce contention for locations. Nonetheless, there can be contention for locations such as the suburb entry location (1 Out Ave, at the top left) and the intersections between the streets and Back Ave (in this suburb, 2, 5, 8, and 11 Back Ave at the right). In addition, one drone may hold up others behind it due to delivering or other delays. For example, in Figure 2 (lower left) D9 will be waiting for D8 to complete its delivery.

The system allows for heavy and fragile parcels. However, the current system handles all parcels the same way, no matter how heavy or whether they are fragile.

The system logs the activity in all locations and various other aspects of the simulation behaviour and generates statistics such as average delivery time.

The simulation operates through time-stepping, that is, each step represents a period of time (actually the standard time taken for a drone to move from one location to the next). In each time-step, every active entity (the drones and the dispatch centre in this case) have the opportunity (they are “ticked”) to take their turn and perform an action in that timestep. Drones cycle through the activities of: (1) waiting for dispatch, (2) transiting from dispatch to the suburb entry point, (3) transiting to delivery address, (4) delivering, (5) transiting to the suburb exit point, (6) transiting back to the dispatch centre, (7) recharge.

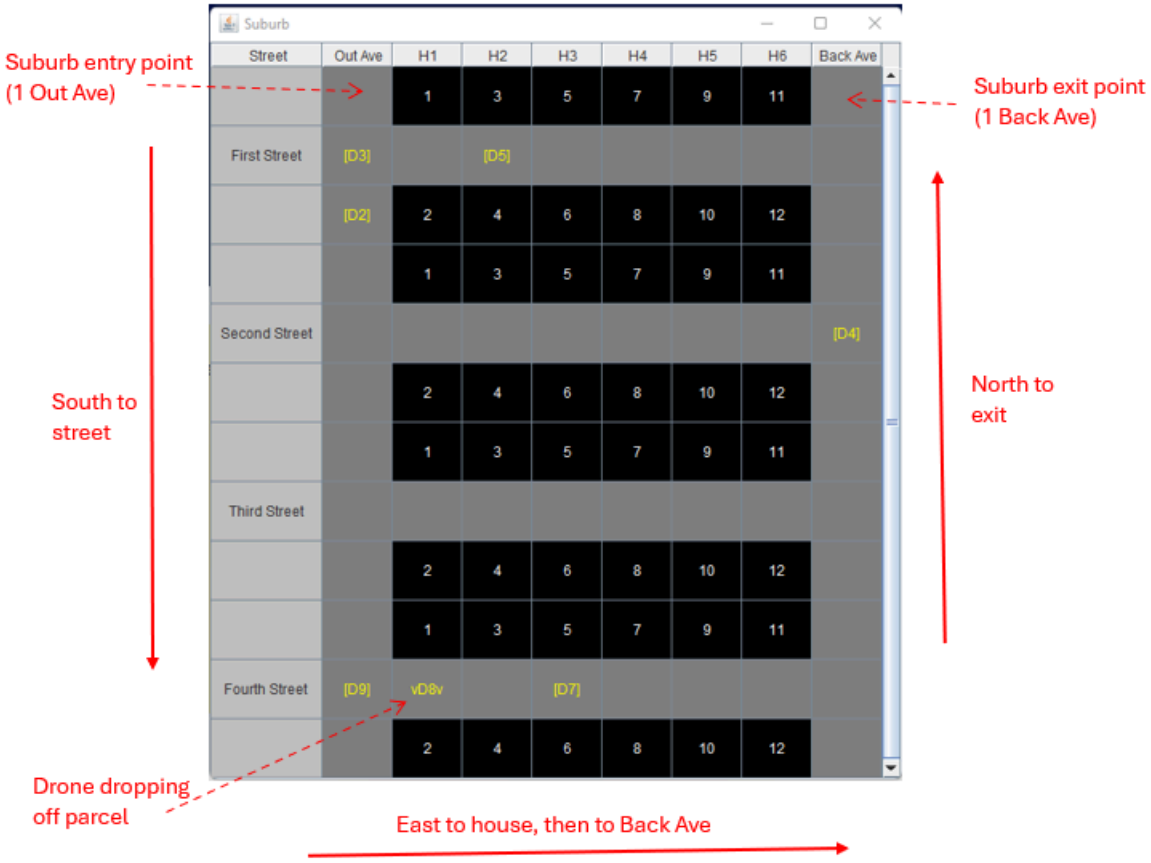


*Figure 2: Droning On Suburb Layout (4 streets by 6 houses) with Drones*

**Requirements for Improvement to the System**

DS are seeking three changes to the system.

*(1) Handling of fragile parcels*

Drone tests on the current delivery approach have shown that the incidence of breakage of fragile parcels is too high. The tests have suggested an alternative, acceptable-breakage-level approach. In this approach, a drone delivering a fragile parcel travels at half speed until it has delivered the parcel before returning to normal speed for the trip back. This impacts on (a) the transit out to the suburb, (b) the transit to the delivery location, and (c) the start of the delivery process (parcel drop-off). This is to be simulated by having the drone alternate between not moving on the first tick and moving on the second tick during (a), (b), and (c).

*(2) Better access to contended locations*

As noted above, a request to move to a location is currently granted to the first requesting drone, and only if that location is currently free. This means that, a request for a location which will become free later in the same cycle will not be granted, and that there is no opportunity to consider all drones seeking to move to a location and choose the highest priority one (see 3 below).

DS is asking that the system be able to:

A. Cache all requests to move to a location made in a time-step.
B. Consider the locations in reverse trip order (clockwise) to increase the opportunity for a drone to be able to move into a location being vacated in the same time-step.

*(3) Priority access to contended locations*

Drones carrying heavy parcels have a more limited range than drones carrying lighter parcels, particularly above a certain weight threshold. Drones carrying fragile parcels will now travel for a longer time. DS would like to be able to give priority to these drones to reduce the risk of drones being unable to deliver/return.

Granting access to the highest priority request for a specific location. The current priority order is as follows:

A. Drones delivering a fragile parcel (whether or not they have dropped it off).
B. Drones delivering a heavy parcel (whether or not they have dropped it off).
C. Drones who are in the location to the south of the requested location (they have travelled further).
D. Any other drones.

If there is more than one drone in the highest priority category, the first request in that category should be granted access.

Note that DS may wish to change the prioritisation in the future, and perhaps eventually even configure it from a properties file (not required for this project). You should take this into account for your choice of design.

**Your Task**

Your task is to consider design options for each of (1), (2) and (3) and implement your best option for each.

**The Base Package**

You will be provided with the source code for the current version of the system, including an example property file. This provides the basis for you to implement the additions described above.

Please carefully study the provided code and ensure that you are confident you understand how it is set up and functions before continuing. Note that you do not need to understand all aspects, just those relevant to the changes you need to make. You should understand the state machine underlying the drone behaviour, and perhaps even model it to help with your work. If you have any questions, please make use of the discussion board.

**Note:** The simulation will run and generate parcels at random times and with random weights and fragility, based on a seed. You can configure the parameters in the property file (*test.properties* by default). Defaults are provided for all parameters in the code.

**Configuration and Project Deliverables**

**(A) Extended Droning On**: As discussed above, and for the users of Droning On to have confidence that changes have been made in a controlled manner, you are required to preserve the Droning On simulation's existing behaviour. Your extended design and implementation must account for the following:

- Preserve the existing behaviour of the system for configurations where there are no fragile or heavy parcels delivered and no contention for locations (e.g. only one drone or sparse parcel arrivals). Note that "preserve" implies identical output. We will use a file comparison tool to check this. Note the simulation default settings (with empty property file) provides such an example.
- Add the above requested changes (1), (2) and (3) to the simulation behaviour.

It's recommended that you understand the high-level design of current system so that you can effectively identify and update relevant parts. You don't need to refactor the whole system, just those parts necessary or helpful to making the required changes.

**(B) Report:** In addition to the extended Droning On system, *DS* also wants you to provide a report to document your design changes and justify your design. Your report should include:

- For each of the changes (1) and (3), a description of your design and one plausible alternative design which you considered, and reasons why your final choice is better.
- a design class diagram which shows all the changed design elements in your submission. It should show how elements you have added relate to provided elements (e.g. associations you have added) but doesn't need to show all the unchanged elements.

More detail of the report is available on the LMS submission page.

**Note:** Your implementation must not violate the principle of the simulation, for example by jumping objects from location to non-adjacent location or having an entity doing more in a time-step than matches the simulation model. We also reserve the right to award or deduct marks for clever or very poor code quality on a case-by-case basis outside of the prescribed marking scheme.

**Testing Your Solution**

We will be testing your application programmatically, so we need to be able to build and run your program without using an integrated development environment. *The entry point must remain as "Main.main()". You must not change the names of properties in the provided property file or require the presence of additional properties. You should not change the graphics (SuburbView) or logging (Logger).*

**Note:** It is **your team's responsibility** to ensure that the team has thoroughly tested their software before submission.

Here are sample outputs from the original system for each of:

- **built-in default parameters (https://canvas.lms.unimelb.edu.au/courses/211063/files/22653600?wrap=1) (https://canvas.lms.unimelb.edu.au/courses/211063/files/22653600/download?download_frd=1)**
- **test.properties (https://canvas.lms.unimelb.edu.au/courses/211063/files/22653890?wrap=1) (https://canvas.lms.unimelb.edu.au/courses/211063/files/22653890/download?download_frd=1)**

Here are sample outputs for the requested system:

- running with the built-in parameters produces the same output as the original system
- **test.properties (https://canvas.lms.unimelb.edu.au/courses/211063/files/22653950?wrap=1) (https://canvas.lms.unimelb.edu.au/courses/211063/files/22653950/download?download_frd=1)** (here is **video (https://canvas.lms.unimelb.edu.au/media_attachments_iframe/22653956)** of the associated behaviour)
- Some higher contention examples:
  - No fragile items: **test_contention.properties (https://canvas.lms.unimelb.edu.au/courses/211063/files/22689473?wrap=1) (https://canvas.lms.unimelb.edu.au/courses/211063/files/22689473/download?download_frd=1)** , **logfile_contention.txt (https://canvas.lms.unimelb.edu.au/courses/211063/files/22689470?wrap=1) (https://canvas.lms.unimelb.edu.au/courses/211063/files/22689470/download?download_frd=1)**
  - With fragile: **test_fragile.properties (https://canvas.lms.unimelb.edu.au/courses/211063/files/22689507?wrap=1) (https://canvas.lms.unimelb.edu.au/courses/211063/files/22689507/download?download_frd=1)** , **logfile_fragile.txt (https://canvas.lms.unimelb.edu.au/courses/211063/files/22689504?wrap=1) (https://canvas.lms.unimelb.edu.au/courses/211063/files/22689504/download?download_frd=1)**

### Submission

Detailed submission instructions will be posted on the LMS. *You must include your team number in all your pdf submissions, and as a comment in all changed or new source code files provided as part of your submission.*